PORTFOLIO PROJECT

Telco Customer Churn Analysis

Turning raw churn data into actionable business insights

26.5%	$139K+	3 Months
Overall Churn Rate	Monthly Revenue at Risk	Peak Churn Window

BUSINESS CONTEXT

Why Churn Analysis Matters

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Revenue Impact

Every churned customer represents lost recurring revenue. Retaining existing customers costs 5× less than acquiring new ones.

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Business Question

Which customers are most likely to churn — and what actions can reduce that risk before it's too late?

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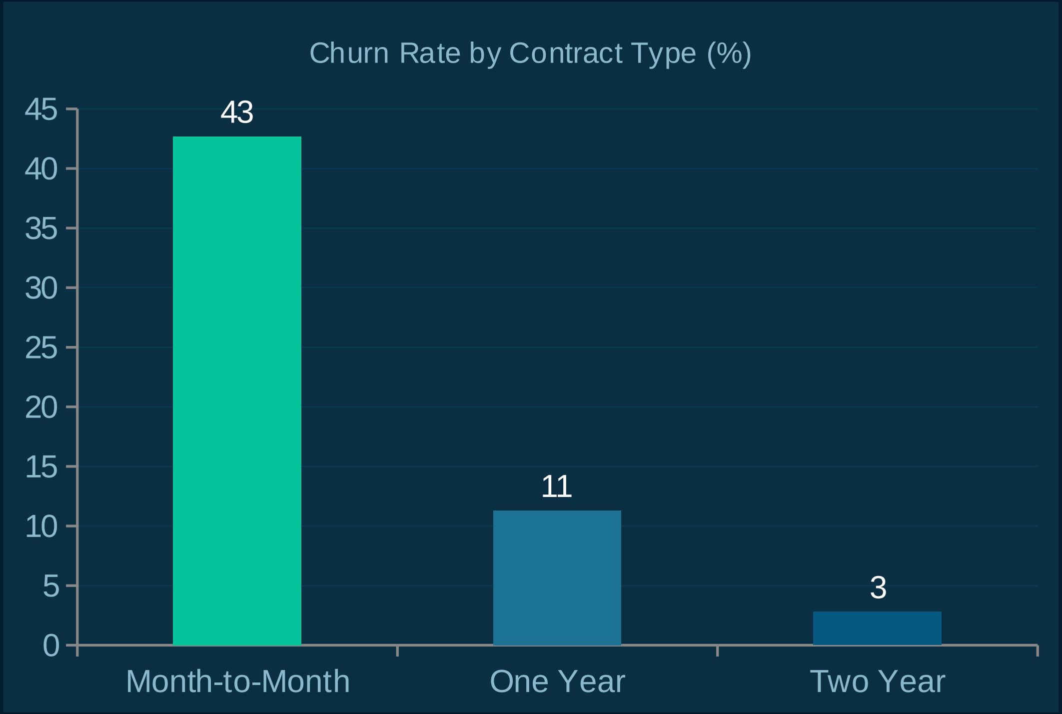
Our Approach

Segment customers by contract, payment method, tenure, and service type to find patterns that drive churn.

KEY FINDINGS

What the Data Reveals



42.7%

Month-to-month churn rate — 15× higher than two-year contracts

45%

Electronic check users churn at highest rate of any payment method

6+ mo

Customer tenure before churn risk drops below average

REVENUE IMPACT

Quantifying the Cost of Churn

$139,130

Monthly revenue lost
to churned customers

Avg. $65/month per churned customer

1,869

Total churned customers

26.5%

Overall churn rate

18 mo

Avg. tenure of churned customers vs 37 mo for retained

RECOMMENDATIONS

3 Actionable Steps to Reduce Churn

01

Push Annual Contracts

Month-to-month churn is 15× higher than two-year plans. Offer a 10–15% discount to convert monthly customers to annual — ROI turns positive within 3 months.

02

Convert Electronic Check Users

Electronic check customers churn at 45%. Incentivize auto-pay enrollment with a small monthly credit. Removes friction and signals commitment.

03

New Customer Early Engagement

Churn spikes in months 1–3. Trigger a proactive outreach sequence at Day 7, Day 30, and Day 90 to address concerns before cancellation decisions are made.

TOOLS USED

- Python (Pandas, Seaborn, Matplotlib)
- Jupyter Notebook
- IBM Telco Dataset (Kaggle)
- Data cleaning module (src/telco_churn)

Have a dataset and a business question? Let's talk.

One-time delivery · No maintenance required · Results in plain English